NOTATION DEFINITION KEY EQUATION SHORTHAND ALGORITHM THEOREM

1 Notation and Symbols

1.1 Variables and Sets NOTATION

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t	Time step index ($t = 0, 1, 2, \ldots$)
S_t, s, s'	State (RV), particular values; s' = next state
A_t, a, a'	Action (RV), particular values
R_t, r	Reward (RV), particular value
G_t	Return (cumulative discounted reward) from t
T	Terminal time step (episodic tasks)
γ	Discount factor, $0 \leq \gamma \leq 1$
α	Step-size / learning rate
θ	Convergence threshold
Δ	Max value change across a sweep
k	Iteration index
$\mathcal{S}, \mathcal{S}^+$	State set; with terminal state(s)
$\mathcal{A}, \mathcal{A}(s)$	Action set; actions available in s
$\mathcal{R}$	Reward set, $\mathcal{R} \subset \mathbb{R}$

1.2 Policies NOTATION

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π	Policy (states $\to$ action probabilities)
$\pi(a \mid s)$	Prob. of action a in state s under π
$\pi(s)$	Deterministic policy action in s
π_*	Optimal policy
π'	Improved policy

1.3 Value Functions NOTATION

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$v_\pi(s)$	State-value: expected return from s under π
$q_\pi(s,a)$	Action-value: expected return from s, taking a, under π
$v_*(s)$	Optimal state-value: $\max_\pi v_\pi(s)$
$q_*(s,a)$	Optimal action-value: $\max_\pi q_\pi(s,a)$
$V(s), V_k(s)$	Estimate of value function; at iteration k

1.4 Probability and Dynamics NOTATION

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$\mathbb{E}_\pi[\cdot]$	Expected value under policy π
$p(s', r \mid s, a)$	Dynamics: $\Pr\{S_t = s', R_t = r \mid S_{t-1} = s, A_{t-1} = a\}$
$p(s' \mid s, a)$	State-transition prob (marginalized over r)
$r(s,a)$	Expected reward for state-action pair
$r(s,a,s')$	Expected reward for state-action-next-state

1.5 Operators SHORTHAND

Horseshoec

$\sum_a$	Shorthand for $\sum_{a \in \mathcal{A}(s)}$
$\sum_{s',r}$	Shorthand for $\sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}}$
$\max_a$	Maximize over $a \in \mathcal{A}(s)$
$\arg\max_a$	Action achieving the maximum
$\leftarrow$	Assignment (update stored value)
$\geq$ (policies)	$\pi \geq \pi'$: $v_\pi(s) \geq v_{\pi'}(s)$ for all s

2 Temporal-Difference Learning ALGORITHM

$$V(S_t) \leftarrow V(S_t) + \alpha\left[V(S_{t+1}) - V(S_t)\right]$$

Combines Monte Carlo (learning from experience) and DP (bootstrapping from estimates).

3 Finite MDPs

3.1 Agent-Environment Interface DEFINITION

Trajectory: $S_0, A_0, R_1, S_1, A_1, R_2, S_2, A_2, R_3, \ldots$

At each t: observe $S_t \in \mathcal{S}$, select $A_t \in \mathcal{A}(s)$, receive $R_{t+1} \in \mathcal{R}$, transition to S_{t+1}.

3.2 Dynamics Function DEFINITION

$$p(s', r|s, a) = \Pr\{S_t = s', R_t = r \mid S_{t-1} = s, A_{t-1} = a\}$$

Must satisfy: $\sum_{s'} \sum_r p(s', r|s,a) = 1$ for all s, a.

3.3 Derived Quantities SHORTHAND

$$p(s'|s,a) = \sum_r p(s', r|s,a)$$

$$r(s,a) = \sum_r r \sum_{s'} p(s', r|s,a)$$

$$r(s,a,s') = \sum_r r \frac{p(s', r|s,a)}{p(s'|s,a)}$$

3.4 Markov Property THEOREM

Future depends only on current state:

$$\Pr\{S_t = s', R_t = r \mid S_0, A_0, \ldots, S_{t-1}, A_{t-1}\} = p(s', r \mid S_{t-1}, A_{t-1})$$

3.5 Returns DEFINITION

Episodic: $G_t = R_{t+1} + R_{t+2} + \cdots + R_T$

Continuing (discounted): $G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$

Recursive: $G_t = R_{t+1} + \gamma G_{t+1}$

Constant reward: $G_t = \sum_{k=0}^{\infty} \gamma^k = \frac{1}{1-\gamma}$

3.6 Policies DEFINITION

$\pi(a|s) = \Pr\{A_t = a \mid S_t = s\}$

Deterministic: $\pi(s) = a$.

3.7 Value Functions DEFINITION

$$v_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$$

$$q_\pi(s,a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]$$

$$v_\pi(s) = \textstyle\sum_a \pi(a|s)\, q_\pi(s,a)$$

3.8 Bellman Equations KEY EQUATION

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s',r} p(s', r|s,a)\left[r + \gamma v_\pi(s')\right]$$

$$q_\pi(s,a) = \sum_{s',r} p(s', r|s,a)\left[r + \gamma \sum_{a'} \pi(a'|s') q_\pi(s', a')\right]$$

3.9 Optimal Value Functions DEFINITION

$$v_*(s) = \max_\pi v_\pi(s) \qquad q_*(s,a) = \max_\pi q_\pi(s,a)$$

$$v_*(s) = \max_a q_*(s,a) \qquad q_*(s,a) = \sum_{s',r} p(s', r|s,a)\left[r + \gamma v_*(s')\right]$$

3.10 Bellman Optimality Equations KEY EQUATION

$$v_*(s) = \max_a \sum_{s',r} p(s', r|s,a)\left[r + \gamma v_*(s')\right]$$

$$q_*(s,a) = \sum_{s',r} p(s', r|s,a)\left[r + \gamma \max_{a'} q_*(s', a')\right]$$

3.11 Optimal Policy DEFINITION

$$\begin{aligned} \pi_*(s) &= \arg\max_a q_*(s,a) \\ &= \arg\max_a \sum_{s',r} p(s', r|s,a)\left[r + \gamma v_*(s')\right] \end{aligned}$$

Any policy greedy w.r.t. v_* or q_* is optimal.

4 Dynamic Programming

DP requires perfect model: $p(s', r|s,a)$.

4.1 Policy Evaluation (Prediction) ALGORITHM

Goal: compute v_π for given π. Iterative update:

$$V_{k+1}(s) = \sum_a \pi(a|s) \sum_{s',r} p(s', r|s,a)\left[r + \gamma V_k(s')\right]$$

Stop when $\max_s |V_{k+1}(s) - V_k(s)| < \theta$. As $k \to \infty$, $V_k \to v_\pi$.